KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · MATH

Apple Harvest in the Blue Ridge — Teacher Lesson Plan & Answer Key

Learning objectives: Students divide a unit fraction by a whole number (sharing: 1/2 ÷ 3 = 1/6) and a whole number by a unit fraction (how many fit: 3 ÷ 1/4 = 12) using tape diagrams and number lines, decide which kind of division a story needs, write and solve the equation, check every quotient with multiplication, and solve two-step problems that convert a quotient to ounces or minutes.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Silas at his family's apple orchard near Hendersonville, North Carolina, with Grandma Lucinda (farm stand), Mr. Barrera (picking crew and apple chips) and Ms. Dunbar (baker). The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Multiplying Fractions (Unit 3 Review) (4 pts, 2 each). If a student misses both warm-up slides, rework one product on paper before moving on.
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): two more typed products; the last one bridges to division.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W16.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Apple Country in the Mountains: nonfiction connection about Henderson County, which grows most of North Carolina's apples. No questions.
6. Slide 5 — Teach: Sharing a Fraction (unit fraction ÷ whole number). Walk through the leftover apple crisp: 1/2 of a pan shared by 3 pickers is 1/6 of the whole pan. Stress: sharing a unit fraction makes a SMALLER piece.
7. Slide 6 — Your Turn, Part 1: Unit Fraction ÷ Whole Number (8 pts, 4 each).
8. Slide 7 — Your Turn, Part 2 (4 pts): one more sharing problem.
9. Slide 8 — Teach: How Many Fit? (whole number ÷ unit fraction). Mr. Barrera's 3 pounds of apple chips in 1/4-pound bags fill 12 bags. Use the number line: count the jumps of 1/4.
10. Slide 9 — Your Turn, Part 1: Whole Number ÷ Unit Fraction (8 pts, 4 each).
11. Slide 10 — Your Turn, Part 2 (4 pts): one more problem.
12. Slide 11 — Teach: Which Kind of Division Is It? Grandma Lucinda's two questions: "How much does each person get?" and "How many can we make?" Have the student name the question type before writing the equation.
13. Slide 12 — Your Turn, Part 1: Choose, Write, and Solve (8 pts, 4 each).
14. Slide 13 — Your Turn, Part 2 (4 pts): one more problem, which may be a missing-number form (for example, 1/4 ÷ ___ = 1/20).
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (unit fraction, dividend, divisor, quotient, inverse operations, tape diagram, number line, denominator).

16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Two-Step Problems: Divide, Then Keep Going. A quotient like 1/8 pound becomes 2 ounces; 1/4 hour becomes 15 minutes.
18. Slide 17 — Your Turn: Divide, Then Convert (6 pts, 3 each).
19. Slide 18 — Real World: Bags, Jars, and Bottles (4 pts): a different situation in each set (apple chip bags, apple butter jars, cider bottles): divide, then subtract.
20. Slide 19 — Real World: The Bakery Counter (4 pts): share a leftover fraction of a cake, crisp or pie, then count pieces.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a “which explanation is correct?” item and a “find the mistake” item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences, same rules. Both serial-comma and no-serial-comma forms of a list are accepted where a list appears.
24. Slide 23 — Match-Up Round (10 pts): 5 division problems matched to quotients, 2 pts each on the first try. Watch the order of the numbers (1/2 ÷ 4 is not 4 ÷ 1/2).
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student’s own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: Apple Festival Day (8 pts, 4 each): a different festival-stand situation in each set, then two-step questions (for example, how many dollars the stand earns, or how many pounds of pie each judge gets).
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W16.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Apple Harvest Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Blue Ridge Fraction Division Master!” with the congratulations message “You helped the whole orchard get ready for the Apple Festival!” and a big confetti burst; below 85 shows “Good Work in the Orchard!” with encouragement (“You are getting closer every time.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: MULTIPLYING FRACTIONS (UNIT 3 REVIEW) (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: A whole number × a fraction: type the product. 6 × 2/3 =	**4** (also accepted: 12/3, 4/1)	6 × 2/3 = 12/3, and 12 ÷ 3 = 4.
r2_s0	Set 1: A fraction × a fraction: 3/5 × 1/2 =? [3/10 / 4/7 / 3/7 / 4/10]	**3/10**	Multiply the numerators (3 × 1 = 3) and multiply the denominators (5 × 2 = 10). The product is 3/10.
r3_s0	Set 1: Type the product as a fraction or a mixed number. 3/8 × 4 =	**1 1/2** (also accepted: 3/2, 12/8, 6/4, 1.5)	3/8 × 4 = 12/8. Simplify: 12/8 = 3/2 = 1 1/2.

ID	Question	Answer	Why (explain to student)
r4_s0	Set 1: Bridge to today: type the product. 1/2 × 1/4 =	**1/8**	1 × 1 = 1 and 2 × 4 = 8, so 1/2 × 1/4 = 1/8. Taking 1/4 of a half gives 1/8. Remember this: today you will see that 1/2 ÷ 4 also equals 1/8.
r1_s1	Set 2: A whole number × a fraction: type the product. 15 × 2/5 =	**6** (also accepted: 30/5, 6/1)	15 × 2/5 = 30/5, and 30 ÷ 5 = 6.
r2_s1	Set 2: A fraction × a fraction: 2/3 × 1/4 =? (simplest form) [1/6 / 3/7 / 2/7 / 3/12]	**1/6**	Multiply the numerators (2 × 1 = 2) and the denominators (3 × 4 = 12) to get 2/12. Simplify by dividing both by 2: 1/6.
r3_s1	Set 2: Type the product as a fraction or a mixed number. 5/6 × 3 =	**2 1/2** (also accepted: 5/2, 15/6, 10/4, 2.5)	5/6 × 3 = 15/6. Simplify: 15/6 = 5/2 = 2 1/2.
r4_s1	Set 2: Bridge to today: type the product. 1/3 × 1/5 =	**1/15**	1 × 1 = 1 and 3 × 5 = 15, so 1/3 × 1/5 = 1/15. Remember this: today you will see that 1/3 ÷ 5 also equals 1/15.
r1_s2	Set 3: A whole number × a fraction: type the product. 12 × 3/4 =	**9** (also accepted: 36/4, 9/1)	12 × 3/4 = 36/4, and 36 ÷ 4 = 9.
r2_s2	Set 3: A fraction × a fraction: 3/4 × 1/3 =? (simplest form) [1/4 / 4/7 / 3/7 / 4/12]	**1/4**	Multiply the numerators (3 × 1 = 3) and the denominators (4 × 3 = 12) to get 3/12. Simplify by dividing both by 3: 1/4.
r3_s2	Set 3: Type the product as a fraction or a mixed number. 2/9 × 6 =	**1 1/3** (also accepted: 4/3, 12/9, 8/6)	2/9 × 6 = 12/9. Simplify: 12/9 = 4/3 = 1 1/3.
r4_s2	Set 3: Bridge to today: type the product. 1/4 × 1/6 =	**1/24**	1 × 1 = 1 and 4 × 6 = 24, so 1/4 × 1/6 = 1/24. Remember this: today you will see that 1/4 ÷ 6 also equals 1/24.

YOUR TURN: UNIT FRACTION ÷ WHOLE NUMBER (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Ms. Dunbar has 1/3 of a pan of apple bread left. She cuts it into 4 equal pieces. What fraction of the WHOLE pan is each piece? 1/3 ÷ 4 =	**1/12** (also accepted: 1/12 of the pan, 1/12 pan, 1/12 of a pan)	1/3 ÷ 4 = 1/(3 × 4) = 1/12. Check: 1/12 × 4 = 4/12 = 1/3.
l2_s0	Set 1: Which multiplication equation is true AND checks that 1/5 ÷ 2 = 1/10? [1/10 × 2 = 1/5 / 1/10 × 2 = 1/20 / 1/5 × 1/2 = 1/7 / 2 × 1/5 = 1/10]	**1/10 × 2 = 1/5**	To check division, multiply the quotient by the divisor. You should get back the number you started with: 1/10 × 2 = 2/10 = 1/5. The other three equations are false.
l3_s0	Set 1: Silas pours 1/2 gallon of apple cider equally into 6 cups. How much of a gallon is in each cup? Gallon in each cup:	**1/12** (also accepted: 1/12 gallon, 1/12 gal, 1/12 of a gallon, 1/12 gallons)	1/2 ÷ 6 = 1/(2 × 6) = 1/12 gallon. Check: 1/12 × 6 = 6/12 = 1/2.
l1_s1	Set 2: There is 1/4 of a pan of apple crisp left. Silas shares it equally among 5 pickers. What fraction of the WHOLE pan does each picker get? 1/4 ÷ 5 =	**1/20** (also accepted: 1/20 of the pan, 1/20 pan, 1/20 of a pan)	1/4 ÷ 5 = 1/(4 × 5) = 1/20. Check: 1/20 × 5 = 5/20 = 1/4.

ID	Question	Answer	Why (explain to student)
l2_s1	Set 2: Which multiplication equation is true AND checks that 1/3 ÷ 6 = 1/18? [1/18 × 6 = 1/3 / 1/18 × 6 = 1/12 / 1/3 × 1/6 = 1/9 / 6 × 1/3 = 1/18]	**1/18 × 6 = 1/3**	To check division, multiply the quotient by the divisor: 1/18 × 6 = 6/18 = 1/3. That is the number you started with. The other three equations are false.
l3_s1	Set 2: Mr. Barrera spreads 1/3 pound of grass seed equally over 2 bare patches between the tree rows. How much of a pound goes on each patch? Pound on each patch:	**1/6** (also accepted: 1/6 pound, 1/6 lb, 1/6 pounds, 1/6 of a pound)	1/3 ÷ 2 = 1/(3 × 2) = 1/6 pound. Check: 1/6 × 2 = 2/6 = 1/3.
l1_s2	Set 3: Ms. Dunbar has 1/6 of a pan of apple bread left. She cuts it into 3 equal pieces. What fraction of the WHOLE pan is each piece? 1/6 ÷ 3 =	**1/18** (also accepted: 1/18 of the pan, 1/18 pan, 1/18 of a pan)	1/6 ÷ 3 = 1/(6 × 3) = 1/18. Check: 1/18 × 3 = 3/18 = 1/6.
l2_s2	Set 3: Which multiplication equation is true AND checks that 1/8 ÷ 4 = 1/32? [1/32 × 4 = 1/8 / 1/32 × 4 = 1/28 / 1/8 × 1/4 = 1/12 / 4 × 1/8 = 1/32]	**1/32 × 4 = 1/8**	To check division, multiply the quotient by the divisor: 1/32 × 4 = 4/32 = 1/8. That is the number you started with. The other three equations are false.
l3_s2	Set 3: Ms. Dunbar has 1/2 pound of butter. She uses the same amount in each of 4 batches of apple bread. How much of a pound goes in each batch? Pound in each batch:	**1/8** (also accepted: 1/8 pound, 1/8 lb, 1/8 pounds, 1/8 of a pound)	1/2 ÷ 4 = 1/(2 × 4) = 1/8 pound. Check: 1/8 × 4 = 4/8 = 1/2.

YOUR TURN: WHOLE NUMBER ÷ UNIT FRACTION (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Mr. Barrera has 5 pounds of dried apple chips. He packs them into 1/4-pound bags. How many bags can he fill? 5 ÷ 1/4 =	**20** (also accepted: 20 bags, 20 bag)	Each pound fills 4 quarter-pound bags: 5 ÷ 1/4 = 5 × 4 = 20 bags. Check: 20 × 1/4 = 20/4 = 5.
k2_s0	Set 1: Which statement about 6 ÷ 1/2 is true? [It equals 12, because each whole has 2 halves / It equals 3, because dividing always makes a number smaller / It equals 1/12, because 1 ÷ (6 × 2) / It equals 6 1/2, because you add the 1/2]	**It equals 12, because each whole has 2 halves**	6 ÷ 1/2 asks how many halves fit in 6. Each whole has 2 halves, so 6 × 2 = 12. Dividing by a unit fraction gives a BIGGER number.
k3_s0	Set 1: The road around the orchard is 4 miles long. A sign stands at every 1/3 mile, which splits the road into equal 1/3-mile sections. How many sections are there? Sections:	**12** (also accepted: 12 sections, 12 section)	Each mile has 3 thirds: 4 ÷ 1/3 = 4 × 3 = 12 sections. Check: 12 × 1/3 = 12/3 = 4.
k1_s1	Set 2: Ms. Dunbar has 3 cups of oats for crisp topping. Each small pan of apple crisp uses 1/3 cup. How many small pans can she make? 3 ÷ 1/3 =	**9** (also accepted: 9 pans, 9 pan)	Each cup holds 3 thirds: 3 ÷ 1/3 = 3 × 3 = 9 pans. Check: 9 × 1/3 = 9/3 = 3.
k2_s1	Set 2: Which statement about 8 ÷ 1/4 is true? [It equals 32, because each whole has 4 fourths / It equals 2, because 8 ÷ 4 = 2 / It equals 1/32, because 1 ÷ (8 × 4) / It equals 8 1/4, because you add the 1/4]	**It equals 32, because each whole has 4 fourths**	8 ÷ 1/4 asks how many fourths fit in 8. Each whole has 4 fourths, so 8 × 4 = 32. Dividing by a unit fraction gives a BIGGER number.
k3_s1	Set 2: Grandma Lucinda has 6 yards of ribbon for gift baskets. Each basket gets 1/2 yard. How many baskets can she tie? Baskets:	**12** (also accepted: 12 baskets, 12 basket, 12 pieces)	Each yard holds 2 halves: 6 ÷ 1/2 = 6 × 2 = 12 baskets. Check: 12 × 1/2 = 12/2 = 6.

ID	Question	Answer	Why (explain to student)
k1_s2	Set 3: The cider press makes 7 gallons of cider. Silas pours it into 1/2-gallon jugs. How many jugs does he fill? 7 ÷ 1/2 =	**14** (also accepted: 14 jugs, 14 jug)	Each gallon fills 2 half-gallon jugs: 7 ÷ 1/2 = 7 × 2 = 14 jugs. Check: 14 × 1/2 = 14/2 = 7.
k2_s2	Set 3: Which statement about 5 ÷ 1/6 is true? [It equals 30, because each whole has 6 sixths / It equals 5/6, because 5 ÷ 6 = 5/6 / It equals 1/30, because 1 ÷ (5 × 6) / It equals 11, because 5 + 6 = 11]	**It equals 30, because each whole has 6 sixths**	5 ÷ 1/6 asks how many sixths fit in 5. Each whole has 6 sixths, so 5 × 6 = 30. Dividing by a unit fraction gives a BIGGER number.
k3_s2	Set 3: The picking crew has 2 hours before the truck leaves. Picking one row takes 1/4 hour. How many rows can they pick? Rows:	**8** (also accepted: 8 rows, 8 row)	Each hour holds 4 quarter hours: 2 ÷ 1/4 = 2 × 4 = 8 rows. Check: 8 × 1/4 = 8/4 = 2.

YOUR TURN: CHOOSE, WRITE, AND SOLVE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Two pickers share 1/5 of a bag of apple chips equally. What fraction of the whole bag does each picker get? Each picker gets:	**1/10** (also accepted: 1/10 of the bag, 1/10 bag, 1/10 of a bag)	This is a sharing story: 1/5 ÷ 2 = 1/(5 × 2) = 1/10 of the bag. Check: 1/10 × 2 = 2/10 = 1/5.
e2_s0	Set 1: Which equation matches this story? “Grandma Lucinda has 4 apple pies. She cuts every pie into eighths. How many slices of 1/8 pie does she have?” [4 ÷ 1/8 = 32 / 1/8 ÷ 4 = 1/32 / 4 ÷ 8 = 1/2 / 8 ÷ 1/4 = 32]	**4 ÷ 1/8 = 32**	Grandma Lucinda HAS 4 whole pies, and each slice is 1/8 of a pie. That is a "how many fit?" story: 4 ÷ 1/8 = 4 × 8 = 32 slices.
e3_s0	Set 1: Type the missing divisor. 1/4 ÷ ___ = 1/20	**5**	1/4 ÷? = 1/20 means 4 ×? = 20, so the missing number is 5. Check: 1/20 × 5 = 5/20 = 1/4.
e1_s1	Set 2: Grandma Lucinda splits 1/3 gallon of apple butter equally into 3 jars. How much of a gallon goes in each jar? Each jar gets:	**1/9** (also accepted: 1/9 gallon, 1/9 gal, 1/9 of a gallon, 1/9 gallons)	This is a sharing story: 1/3 ÷ 3 = 1/(3 × 3) = 1/9 gallon. Check: 1/9 × 3 = 3/9 = 1/3.
e2_s1	Set 2: Which equation matches this story? “Mr. Barrera has 1/2 of a crate of apples. He splits it equally into 6 baskets. How much of the crate goes in each basket?” [1/2 ÷ 6 = 1/12 / 6 ÷ 1/2 = 12 / 1/2 × 6 = 3 / 6 ÷ 2 = 3]	**1/2 ÷ 6 = 1/12**	Mr. Barrera HAS 1/2 of a crate and shares it 6 ways. That is a "how much in each share?" story: 1/2 ÷ 6 = 1/12 of a crate.
e3_s1	Set 2: Type the missing dividend. ___ ÷ 1/3 = 15	**5**	? ÷ 1/3 = 15 means? × 3 = 15, so the missing number is 5. Check: 15 × 1/3 = 15/3 = 5.
e1_s2	Set 3: Half of the Honeycrisp block of trees (1/2 of the block) is split equally among 5 pickers. What fraction of the whole block does each picker get? Each picker gets:	**1/10** (also accepted: 1/10 of the block, 1/10 block, 1/10 of a block)	This is a sharing story: 1/2 ÷ 5 = 1/(2 × 5) = 1/10 of the block. Check: 1/10 × 5 = 5/10 = 1/2.
e2_s2	Set 3: Which equation matches this story? “Silas has 3 feet of twine. He cuts it into pieces that are each 1/4 foot long. How many pieces does he get?” [3 ÷ 1/4 = 12 / 1/4 ÷ 3 = 1/12 / 3 × 1/4 = 3/4 / 4 ÷ 3 = 1 1/3]	**3 ÷ 1/4 = 12**	Silas HAS 3 feet of twine, and each piece is 1/4 foot. That is a "how many fit?" story: 3 ÷ 1/4 = 3 × 4 = 12 pieces.
e3_s2	Set 3: Type the missing divisor. 1/6 ÷ ___ = 1/24	**4**	1/6 ÷? = 1/24 means 6 ×? = 24, so the missing number is 4. Check: 1/24 × 4 = 4/24 = 1/6.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: Read the sentence from the lesson: “The quotient, 1/6, tells how much of the pan each worker gets.” What does quotient mean? [The answer to a division problem / The number you divide by / The amount you start with / The top number of a fraction]	**The answer to a division problem**	A quotient is the answer to a division problem. In 1/2 ÷ 3 = 1/6, the quotient is 1/6.
v2_s0	Set 1: Type the word: in 3 ÷ 1/4, the number 3 is the ___, because it is the amount Mr. Barrera started with.	**dividend** (also accepted: the dividend)	The dividend is the amount you start with and divide. In 3 ÷ 1/4, Mr. Barrera started with 3 pounds, so 3 is the dividend.
v1_s1	Set 2: Read the sentence from the lesson: “In 3 ÷ 1/4, the divisor is 1/4, the size of each bag.” What does divisor mean? [The number you divide by / The answer to a division problem / The amount you start with / Any fraction with a numerator of 1]	**The number you divide by**	The divisor is the number you divide by. In 3 ÷ 1/4, you divide by 1/4, the size of each bag.
v2_s1	Set 2: Type the missing word: 1/9 is a ___ fraction because its numerator is 1.	**unit** (also accepted: unit fraction, a unit fraction, a unit)	A unit fraction has a numerator of 1, like 1/9.
v1_s2	Set 3: Read the sentence from the lesson: “Multiplication and division are inverse operations, so 1/6 × 3 = 1/2 checks 1/2 ÷ 3 = 1/6.” What does inverse operations mean? [Operations that undo each other / Operations that always give bigger answers / Operations you can only do with whole numbers / Operations that give the same answer]	**Operations that undo each other**	Inverse operations undo each other. Dividing 1/2 by 3 gives 1/6, and multiplying 1/6 by 3 takes you back to 1/2.
v2_s2	Set 3: Type the word: in 1/5 ÷ 4 = 1/20, the answer 1/20 is called the ___.	**quotient** (also accepted: the quotient)	The quotient is the answer to a division problem. In 1/5 ÷ 4 = 1/20, the quotient is 1/20.

YOUR TURN: DIVIDE, THEN CONVERT (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Two steps: Silas splits 1/2 pound of cinnamon equally into 2 shakers. How many OUNCES go in each shaker? (1 pound = 16 ounces) Ounces:	**4** (also accepted: 4 ounces, 4 ounce, 4 oz)	Step 1: 1/2 ÷ 2 = 1/4 pound per shaker. Step 2: 1/4 of 16 ounces is 16 ÷ 4 = 4 ounces.
t2_s0	Set 1: Two steps: Grandma Lucinda sets aside 1/2 hour to judge 3 pies, with the same amount of time for each pie. How many MINUTES does each pie get? Minutes:	**10** (also accepted: 10 minutes, 10 minute, 10 min)	Step 1: 1/2 ÷ 3 = 1/6 hour per pie. Step 2: 1/6 of 60 minutes is 60 ÷ 6 = 10 minutes.
t1_s1	Set 2: Two steps: Ms. Dunbar splits 1/2 pound of chopped walnuts equally into 8 small bags. How many OUNCES are in each bag? (1 pound = 16 ounces) Ounces:	**1** (also accepted: 1 ounce, 1 ounces, 1 oz)	Step 1: 1/2 ÷ 8 = 1/16 pound per bag. Step 2: 1/16 of 16 ounces is 16 ÷ 16 = 1 ounce.
t2_s1	Set 2: Two steps: Silas has 1/4 hour to do 3 small jobs, with the same amount of time for each job. How many MINUTES does each job get? Minutes:	**5** (also accepted: 5 minutes, 5 minute, 5 min)	Step 1: 1/4 ÷ 3 = 1/12 hour per job. Step 2: 1/12 of 60 minutes is 60 ÷ 12 = 5 minutes.

ID	Question	Answer	Why (explain to student)
t1_s2	Set 3: Two steps: Grandma Lucinda splits 1/4 pound of brown sugar equally among 4 batches of apple butter. How many OUNCES go in each batch? (1 pound = 16 ounces) Ounces:	**1** (also accepted: 1 ounce, 1 ounces, 1 oz)	Step 1: 1/4 ÷ 4 = 1/16 pound per batch. Step 2: 1/16 of 16 ounces is 16 ÷ 16 = 1 ounce.
t2_s2	Set 3: Two steps: Mr. Barrera has 1/2 hour to train 5 new pickers, one at a time, with the same amount of time for each. How many MINUTES does each picker get? Minutes:	**6** (also accepted: 6 minutes, 6 minute, 6 min)	Step 1: 1/2 ÷ 5 = 1/10 hour per picker. Step 2: 1/10 of 60 minutes is 60 ÷ 10 = 6 minutes.

REAL WORLD: BAGS, JARS, AND BOTTLES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Mr. Barrera has 6 pounds of dried apple chips. He packs all of them into 1/4-pound bags. At the morning market, he sells 9 bags.* Two steps: how many bags does Mr. Barrera have left after the morning market? Bags left:	**15** (also accepted: 15 bags, 15 bag)	Step 1: 6 ÷ 1/4 = 6 × 4 = 24 bags. Step 2: 24 − 9 = 15 bags left.
w1_s1	Set 2: *The Situation Grandma Lucinda has 5 quarts of apple butter. She fills 1/2-quart jars with all of it. Then she gives 3 jars to the community bake sale.* Two steps: how many jars does Grandma Lucinda have left? Jars left:	**7** (also accepted: 7 jars, 7 jar)	Step 1: 5 ÷ 1/2 = 5 × 2 = 10 jars. Step 2: 10 − 3 = 7 jars left.
w1_s2	Set 3: *The Situation The cider team has 4 gallons of fresh cider. They pour all of it into 1/3-gallon bottles. Then they set 4 bottles aside for free tasting cups.* Two steps: how many bottles are left to sell? Bottles left:	**8** (also accepted: 8 bottles, 8 bottle)	Step 1: 4 ÷ 1/3 = 4 × 3 = 12 bottles. Step 2: 12 − 4 = 8 bottles left to sell.

REAL WORLD: THE BAKERY COUNTER (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Ms. Dunbar has 1/2 of a large apple cake left. She cuts it into 6 equal pieces. Silas buys 2 of the pieces for his family.* Two steps: what fraction of the WHOLE cake does Silas buy? (Simplest form is best.) Fraction of the cake:	**1/6** (also accepted: 2/12, 1/6 of the cake, 2/12 of the cake)	Step 1: one piece is 1/2 ÷ 6 = 1/12 of the whole cake. Step 2: 2 pieces = 2 × 1/12 = 2/12 = 1/6 of the cake.
w2_s1	Set 2: *The Situation Ms. Dunbar has 1/3 of a pan of apple crisp left. She cuts it into 4 equal pieces. By noon, 3 of the pieces are sold.* Two steps: what fraction of the WHOLE pan is sold by noon? (Simplest form is best.) Fraction of the pan:	**1/4** (also accepted: 3/12, 1/4 of the pan, 3/12 of the pan)	Step 1: one piece is 1/3 ÷ 4 = 1/12 of the whole pan. Step 2: 3 pieces = 3 × 1/12 = 3/12 = 1/4 of the pan.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: *The Situation Ms. Dunbar has 1/2 of an apple pie left. She cuts it into 5 equal slices. Silas and his cousin each eat 1 slice.* Two steps: what fraction of the WHOLE pie do Silas and his cousin eat together? (Simplest form is best.) Fraction of the pie:	**1/5** (also accepted: 2/10, 1/5 of the pie, 2/10 of the pie)	Step 1: one slice is 1/2 ÷ 5 = 1/10 of the whole pie. Step 2: 2 slices = 2 × 1/10 = 2/10 = 1/5 of the pie.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? 1/3 ÷ 2 =? [1/6, because splitting a third into 2 equal parts makes sixths of the whole / 2/3, because 1/3 × 2 = 2/3 / 6, because 3 × 2 = 6 / 1/5, because 3 + 2 = 5]	**1/6, because splitting a third into 2 equal parts makes sixths of the whole**	Sharing 1/3 into 2 equal parts cuts every third of the whole in half, so the whole has 6 equal parts. Each share is 1/6. Check: 1/6 × 2 = 2/6 = 1/3.
x2_s0	Set 1: Find the mistake: Anselm says 4 ÷ 1/2 = 2, because "dividing always makes a number smaller." What went wrong? [He found 4 ÷ 2 instead. There are 2 halves in each whole, so 4 ÷ 1/2 = 8 / Nothing. 4 ÷ 1/2 = 2 / He should get 1/8, because 1 ÷ (4 × 2) = 1/8 / He should get 4 1/2, because you add the 1/2]	**He found 4 ÷ 2 instead. There are 2 halves in each whole, so 4 ÷ 1/2 = 8**	4 ÷ 1/2 asks how many halves fit in 4. Each whole has 2 halves, so 4 × 2 = 8. Dividing by a number less than 1 gives a BIGGER answer, so "dividing always makes smaller" is not true.
x1_s1	Set 2: Which explanation is correct? 5 ÷ 1/3 =? [15, because each whole has 3 thirds, so 5 wholes have 5 × 3 = 15 thirds / 5/3, because 5 ÷ 3 = 5/3 / 1/15, because 1/3 ÷ 5 = 1/15 / 8, because 5 + 3 = 8]	**15, because each whole has 3 thirds, so 5 wholes have 5 × 3 = 15 thirds**	5 ÷ 1/3 asks how many thirds fit in 5. Each whole has 3 thirds, so 5 wholes have 5 × 3 = 15 thirds. Check: 15 × 1/3 = 15/3 = 5.
x2_s1	Set 2: Find the mistake: Esme says 1/4 ÷ 3 = 3/4. What went wrong? [She multiplied the numerator. Sharing 1/4 three ways makes smaller pieces: 1/4 ÷ 3 = 1/12 / Nothing. 1/4 ÷ 3 = 3/4 / She should get 12, because 4 × 3 = 12 / She should get 1/7, because 4 + 3 = 7]	**She multiplied the numerator. Sharing 1/4 three ways makes smaller pieces: 1/4 ÷ 3 = 1/12**	Sharing 1/4 among 3 people makes each share SMALLER than 1/4. Multiply the denominator, not the numerator: 1/4 ÷ 3 = 1/12. 3/4 is bigger than 1/4, so it cannot be right.
x1_s2	Set 3: Which explanation is correct? Which is greater, 1/2 ÷ 4 or 4 ÷ 1/2? [4 ÷ 1/2, because it equals 8, and 1/2 ÷ 4 equals only 1/8 / They are equal, because they use the same numbers / 1/2 ÷ 4, because dividing by 4 makes a bigger number / 4 ÷ 1/2, because it equals 2, and 1/2 ÷ 4 equals 1/8]	**4 ÷ 1/2, because it equals 8, and 1/2 ÷ 4 equals only 1/8**	1/2 ÷ 4 = 1/8 (a half shared 4 ways) and 4 ÷ 1/2 = 8 (halves that fit in 4). 8 is much greater than 1/8. Order matters in division.
x2_s2	Set 3: Find the mistake: Anselm checks 1/5 ÷ 2 = 1/10 by writing 1/10 × 5 = 1/2. What should he fix? [Multiply the quotient by the divisor: 1/10 × 2 = 2/10 = 1/5, which matches the dividend / Nothing. His check is correct / He should check with addition: 1/10 + 2 = 1/5 / 1/5 ÷ 2 is really 2/5, so his answer is wrong]	**Multiply the quotient by the divisor: 1/10 × 2 = 2/10 = 1/5, which matches the dividend**	To check division, multiply the quotient by the DIVISOR: 1/10 × 2 = 2/10 = 1/5, which is the dividend. Anselm multiplied by 5, which does not check this problem, even though 1/10 × 5 = 1/2 is a true fact.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: silas asked, how many quarter-pound bags can we fill	**Silas asked, "How many quarter-pound bags can we fill?"**	Capitalize Silas (the first word and a name), put quotation marks around his exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: Grandma Lucinda and me packed the jars.	**Grandma Lucinda and I packed the jars.**	Use I, not me, when the word is part of the subject. Test it by dropping Grandma Lucinda: you would say "I packed the jars," not "me packed the jars."
s3_s0	Set 1: Fix this sentence: We picked Honeycrisp Gala and Fuji apples.	**We picked Honeycrisp, Gala, and Fuji apples.** (also accepted: We picked Honeycrisp, Gala and Fuji apples.)	Use commas to separate the items in a series: Honeycrisp, Gala, and Fuji.
s4_s0	Set 1: Fix this sentence: the bags of apple chips is ready to sell	**The bags of apple chips are ready to sell.**	Capitalize the first word. The subject is bags (plural), not chips, so use are, not is. End with a period.
s1_s1	Set 2: Fix this sentence: grandma lucinda said, each jar holds half a quart	**Grandma Lucinda said, "Each jar holds half a quart."**	Capitalize Grandma Lucinda (a family title used as a name), put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The pickers left there ladders by the barn.	**The pickers left their ladders by the barn.**	Their shows ownership: the ladders belong to the pickers. There tells a place.
s3_s1	Set 2: Fix this sentence: After lunch Silas divided the cider into jugs.	**After lunch, Silas divided the cider into jugs.**	Put a comma after an introductory phrase that comes before the main part of the sentence: After lunch, Silas divided the cider into jugs.
s4_s1	Set 2: Fix this sentence: mr. barrera and silas was at the market by seven	**Mr. Barrera and Silas were at the market by seven.**	Capitalize the first word and both names. Two people (Mr. Barrera and Silas) make a plural subject, so use were, not was. End with a period.
s1_s2	Set 3: Fix this sentence: ms. dunbar asked, who wants a slice of apple cake	**Ms. Dunbar asked, "Who wants a slice of apple cake?"**	Capitalize Ms. Dunbar, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Each of the baskets have a label.	**Each of the baskets has a label.**	The subject is each (one), not baskets, so use has, not have.
s3_s2	Set 3: Fix this sentence: Yes the answer is 1/12 of the pan.	**Yes, the answer is 1/12 of the pan.**	Put a comma after an introductory word like yes, no, or well: Yes, the answer is 1/12 of the pan.
s4_s2	Set 3: Fix this sentence: theyre counting the rows in the orchard	**They're counting the rows in the orchard.**	Capitalize the first word. They're is a contraction for they are, so it needs an apostrophe. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 1/2 ÷ 4	**1/8**	2 points if matched on the first try.
match_s0_2	Set 1: Match 4 ÷ 1/2	**8**	2 points if matched on the first try.
match_s0_3	Set 1: Match 1/3 ÷ 2	**1/6**	2 points if matched on the first try.
match_s0_4	Set 1: Match 2 ÷ 1/3	**6**	2 points if matched on the first try.
match_s0_5	Set 1: Match 1/5 ÷ 3	**1/15**	2 points if matched on the first try.
match_s1_1	Set 2: Match 1/4 ÷ 3	**1/12**	2 points if matched on the first try.
match_s1_2	Set 2: Match 3 ÷ 1/4	**12**	2 points if matched on the first try.
match_s1_3	Set 2: Match 1/2 ÷ 5	**1/10**	2 points if matched on the first try.
match_s1_4	Set 2: Match 5 ÷ 1/2	**10**	2 points if matched on the first try.
match_s1_5	Set 2: Match 1/7 ÷ 2	**1/14**	2 points if matched on the first try.
match_s2_1	Set 3: Match 1/3 ÷ 3	**1/9**	2 points if matched on the first try.
match_s2_2	Set 3: Match 3 ÷ 1/3	**9**	2 points if matched on the first try.
match_s2_3	Set 3: Match 1/8 ÷ 2	**1/16**	2 points if matched on the first try.
match_s2_4	Set 3: Match 6 ÷ 1/4	**24**	2 points if matched on the first try.
match_s2_5	Set 3: Match 1/5 ÷ 5	**1/25**	2 points if matched on the first try.

BOSS BATTLE: APPLE FESTIVAL DAY (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Silas and Grandma Lucinda Plan the Festival Stand It is the North Carolina Apple Festival. Grandma Lucinda brings 8 pounds of dried apple chips to pack into 1/4-pound bags, and each bag sells for $3. Ms. Dunbar sends 1/2 of a big apple pie for 4 festival judges to share equally. The WHOLE pie weighs 4 pounds.* Two steps: if every bag sells, how many dollars does the stand earn from apple chips? Dollars:	**96** (also accepted: $96, 96 dollars, 96.00, $96.00)	Step 1: 8 ÷ 1/4 = 8 × 4 = 32 bags. Step 2: 32 × $3 = $96.
b2_s0	Set 1: Two steps: how many POUNDS of pie does each judge get? Pounds:	**1/2** (also accepted: 1/2 pound, 1/2 lb, half a pound, half pound, 0.5, 0.5 pound, 2/4, 4/8)	Step 1: each judge gets 1/2 ÷ 4 = 1/8 of the whole pie. Step 2: 1/8 of 4 pounds = 4 ÷ 8 = 1/2 pound.
b1_s1	Set 2: *Silas and Grandma Lucinda Plan the Festival Stand It is the North Carolina Apple Festival. Grandma Lucinda brings 6 quarts of cold cider to pour into 1/2-quart cups, and each cup sells for $2. Ms. Dunbar sends 1/3 of a pan of apple crisp for 2 festival judges to share equally. The WHOLE pan weighs 6 pounds.* Two steps: if every cup sells, how many dollars does the stand earn from cider? Dollars:	**24** (also accepted: $24, 24 dollars, 24.00, $24.00)	Step 1: 6 ÷ 1/2 = 6 × 2 = 12 cups. Step 2: 12 × $2 = $24.

ID	Question	Answer	Why (explain to student)
b2_s1	Set 2: Two steps: how many POUNDS of apple crisp does each judge get? Pounds:	**1** (also accepted: 1 pound, 1 lb, one pound, one)	Step 1: each judge gets 1/3 ÷ 2 = 1/6 of the whole pan. Step 2: 1/6 of 6 pounds = 6 ÷ 6 = 1 pound.
b1_s2	Set 3: *Silas and Grandma Lucinda Plan the Festival Stand It is the North Carolina Apple Festival. Grandma Lucinda brings 5 pounds of dried apple chips to pack into 1/3-pound bags, and each bag sells for $4. Ms. Dunbar sends 1/4 of a big apple sheet cake for 2 festival judges to share equally. The WHOLE sheet cake weighs 16 pounds.* Two steps: if every bag sells, how many dollars does the stand earn from apple chips? Dollars:	**60** (also accepted: $60, 60 dollars, 60.00, $60.00)	Step 1: 5 ÷ 1/3 = 5 × 3 = 15 bags. Step 2: 15 × $4 = $60.
b2_s2	Set 3: Two steps: how many POUNDS of sheet cake does each judge get? Pounds:	**2** (also accepted: 2 pounds, 2 pound, 2 lb, 2 lbs, two pounds)	Step 1: each judge gets 1/4 ÷ 2 = 1/8 of the whole cake. Step 2: 1/8 of 16 pounds = 16 ÷ 8 = 2 pounds.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W16.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 3/4 × 8 and 2/5 × 1/3	a) 24/4 = 6 b) 2/15
2	1/2 of a pan cut into 2 equal pieces	Each piece is 1/4 of the whole pan (1/2 ÷ 2 = 1/4). The drawing shows the whole pan, the half shaded, and the half split in two.
3	4 pounds into 1/2-pound bags	8 bags (4 ÷ 1/2 = 8); the number line shows 8 jumps of 1/2.
4	1/3 of a pie shared by 2 people	smaller than 1/3 (each share is 1/6)
5	Quotient in 12 ÷ 4 = 3	3

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	1/4 pan cut into 3 pieces; two pickers each eat 1	1/4 ÷ 3 = 1/12 per piece; 2 pieces = 2/12 = 1/6 of the whole pan.
2	6 pounds into 1/3-pound bags; 10 sold	6 ÷ 1/3 = 18 bags; 18 − 10 = 8 bags left.
3	Equation for 1/2 gallon split into 5 jars	1/2 ÷ 5 = 1/10
4	Delia writes 1/6 ÷ 2 = 1/3	Sharing 1/6 between 2 makes a smaller piece, but she made it bigger (she divided the denominator). The correct quotient is 1/12. Check: 1/12 × 2 = 2/12 = 1/6.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	1/3 hour for 4 jugs; 6 jugs at that pace	1/3 ÷ 4 = 1/12 hour = 5 minutes per jug (1/3 hour = 20 minutes, 20 ÷ 4 = 5). 6 jugs × 5 minutes = 30 minutes.